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How Reliable Are LLM Evaluations of Microservice Architecture Diagrams?

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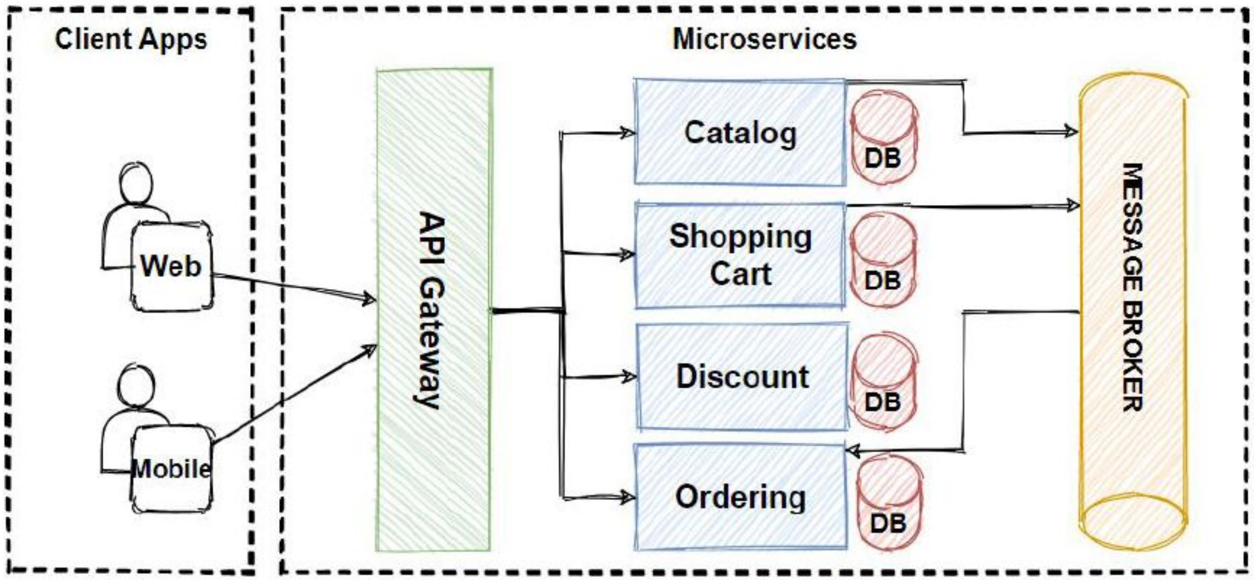
Context & Research Problem

Software Architecture diagrams express service boundaries, API gateways, message brokers, and database ownership in microservices.



Context & Research Problem

- Rapidly evolving systems cause diagrams to become obsolete, incomplete, ambiguous, or overly informal.

- Multimodal LLMs emerge as potential automated reviewers, but their empirical reliability on visual diagrams was unknown.

Background

Baseline Study: Oliveira & Mendonça (ECSA 2025)

"LLM-Based Quality Assessment of Software Architecture Diagrams: A Preliminary Study with Four Open-Source Projects"

Evaluated the feasibility of a single LLM (ChatGPT-4o) on 4 open-source architecture diagrams using 5 criteria (Accuracy, Clarity, Completeness, Consistency, Detail)

Goal and Research Questions

Goal: Evaluate the reliability of LLM-based microservice architecture diagram assessments by analyzing score alignment against human reference scores and expert preferences for generated reasoning.

- **RQ1:** To what extent do LLM-generated scores align with human reference scores when assessing microservice architecture diagrams?
- **RQ2:** Which quality criteria show stronger or weaker agreement between LLM-generated scores and human reference scores?
- **RQ3:** Do software engineers prefer the reasoning generated from image-based or text-based diagram representations?

Goal and Research Questions

1. Data Collection
6 real-world open-source microservice systems. PNG, SVG/XML, and text descriptions.

2. LLM Evaluation
GPT-5 and Claude Sonnet 4.6. 2 representations (PNG vs SVG/XML). 5 criteria, 15 guidelines. 120 API calls

3. Human Audit
16 Software Engineers. 12 balanced human reference sets (ground truth) + 253 pairwise blind justification comparisons.

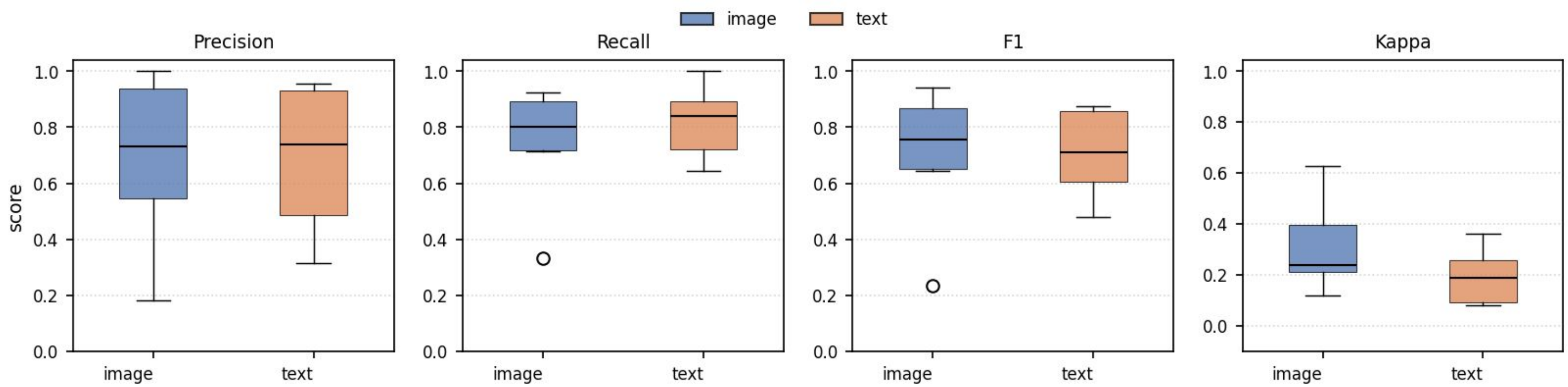
2. Data Analysis
F1, Precision, Recall, Quadratic Weighted Kappa

5 criteria + 15 Guidelines

Quality Criteria	Definition	Guideline-Level Checks
Accuracy	Does the diagram correctly reflect the architecture described in text?	1. Component matching | 2. Interaction matching
Clarity	Can a technical reader easily understand the diagram and relationships?	3. Element identification | 4. Reading flow | 5. Label readability | 6. Specific symbology
Completeness	Can a technical reader easily understand the diagram and relationships?	7. Coverage of main components | 8. External actors coverage | 9. Main interactions coverage
Consistency	Does it apply notation and technology symbols uniformly?	10. Notation definition & uniformity | 11. Technology symbols | 12. Naming consistency
Level of Detail	Is the level of detail appropriate for developers and architects?	13. Element detail | 14. Interaction detail | 15. Overall appropriateness

Scoring scale: 4 = Excellent, 3 = Good, 2 = Fair, 1 = Poor

Results: RQ1



Overall Agreement

The evaluations demonstrate a **reasonable agreement** overall:

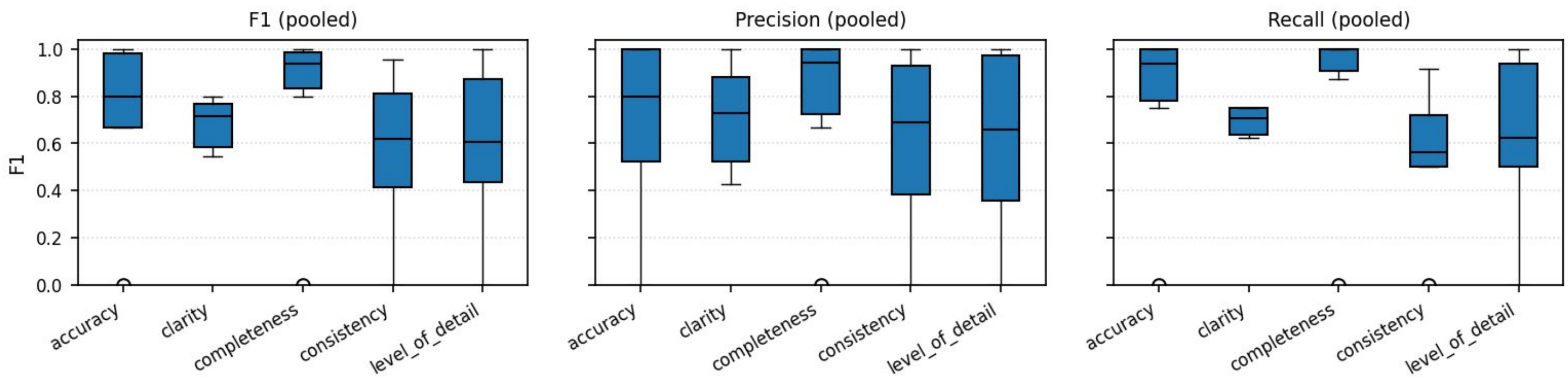
- F1-Score = **0.761**
- Cohen's Kappa (κ) = **0.308**

Images vs. Text

Visual representations descriptively **outperform** text-based descriptions:

- F1-Score: **0.776** vs. 0.747
- Kappa (κ): **0.387** vs. 0.225

Results: RQ2



Information Presence

LLMs excel in **Completeness** (Recall 0.975) and **Accuracy** (Recall 0.938).

Outstanding performance when checking checklist-style component presence/absence.

Subjective Interpretation

LLMs underperform in **Clarity** (lowest F1: 0.697, Recall: 0.679) and **Level of Detail**.

Difficulty evaluating visual layout readability and audience expectations.

Results: RQ3

- Image-based reasoning preferred: 44.7% vs. 28.9% for text; 26.4% ties.
- Among decided comparisons, 60.8% favored images ($p = 0.0034$).
- Image reasoning was preferred in accuracy, clarity, completeness, and consistency.
- Level of detail was the only criterion favoring text.

Preference	Count	Percentage
Image	113	44.7%
Text	73	28.9%
Tie	67	26.4%
Total	253	100.0%

Criterion	Image	Text	Image share
Accuracy	18	15	54.5%
Clarity	29	18	61.7%
Completeness	23	11	67.6%
Consistency	28	13	68.3%
Level of Detail	15	16	48.4%

Key Findings

- LLMs should not act as authoritative evaluators. They function best as first-pass triage filters to flag incomplete or suspect diagrams for human inspection.

- Decomposing evaluation into 15 concrete guidelines converts generic criticisms ("diagram is poor") into actionable, auditable feedback items.

- The optimal workflow combines automated LLM triage with human expert validation, especially for subjective criteria like clarity and level of detail.

Thank you!

Questions?